STATISTICAL LEARNING THEORY (03/03/26)

Summer Term 2026

Other course names:

Machine Learning (Data Science Master)
Adv. Machine Learning (AIS Master)

MOTIVATION

Example:

	$\in \mathbb{R}$ Weight (in g)	$\in \mathbb{R}$ Color (in [0,1])	$\in \{0,1\}$ Tasty?
Papaya 1	800	0.1	0
⋮	⋮	⋮	⋮
Papaya N	1200	0.7	1

0 for TASTY? means "not tasty" and 1 means "tasty".

Lets call the information in this table our training data.

Based on that, we try to find

$$\boxed{h: \mathbb{R} \times \mathbb{R} \to \{0,1\}}$$

i.e., a function that takes a tuple of (weight, color) and outputs a prediction of whether a papaya is tasty (1) or not (0).

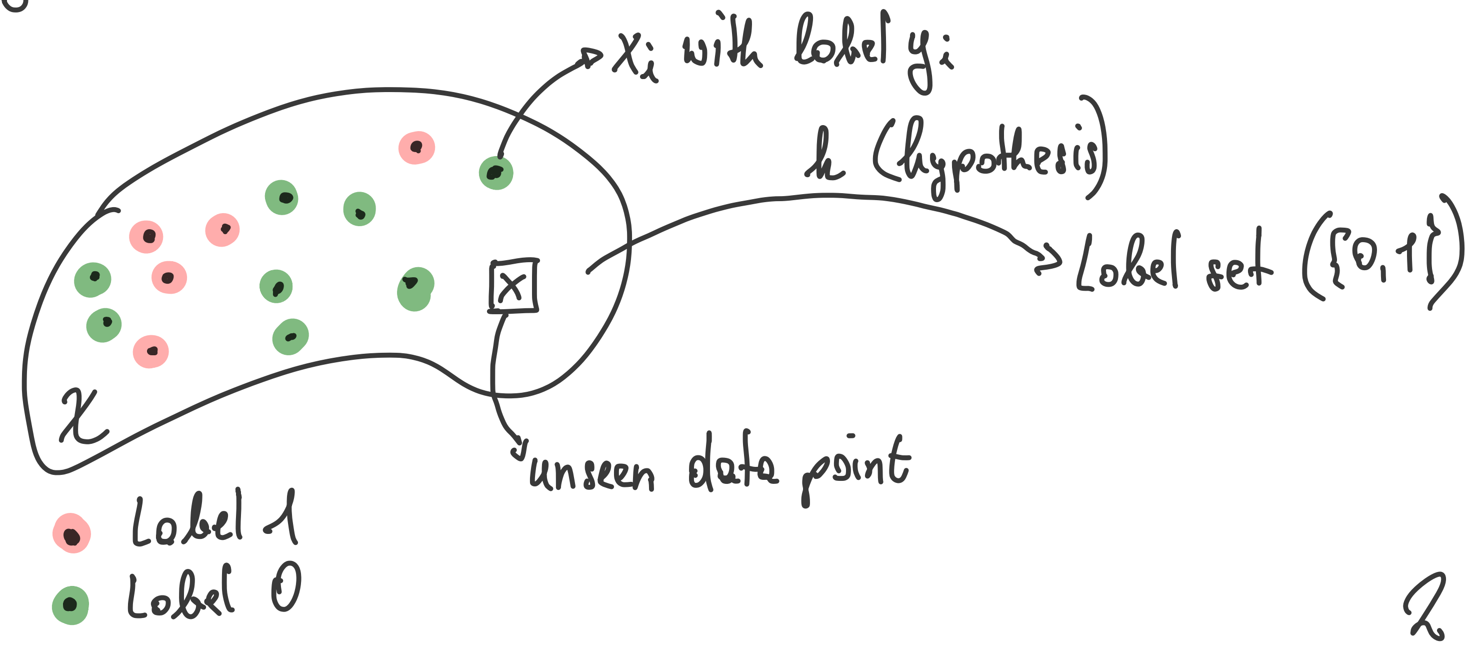
We are going to call such a function (h) a **hypothesis.**

More formally, the available data comes in the form of

$$\big((x_1, y_1), \ldots\ldots, (x_N, y_N)\big) = S$$

with (in our example) $x_i \in \underbrace{\mathbb{R}^2}_{\text{domain } \mathcal{X}}$ and $y_i \in \underbrace{\{0,1\}}_{\text{Label set } \mathcal{Y}}$.

We say a **learner** (some algorithm) receives the training data S and returns h!



Two assumptions we are going to make initially:

A.1. All the x_i's are drawn <u>independently and identically distributed</u> (iid) from some (unknown) distribution D over our domain $\mathcal{X}$.

A.2. The x_i's are labeled by some (unknown) function

$$\boxed{f: \mathcal{X} \to \mathcal{Y}}$$

called the true labeling function.

This means our training data is of the form

$$S = \Big((x_1, \underbrace{f(x_1)}_{y_1}), \ldots\ldots, (x_N, \underbrace{f(x_N)}_{y_N}) \Big)$$

<u>What do we care about?</u>

true labeling function

$$\{ x \in \mathcal{X} : h(x) \neq f(x) \} = A$$

hypothesis

i.e., all points in our domain $\mathcal{X}$ for which h <u>differs</u> from f.

Specifically, we are interested in the probability of making a mistake $(h(x) \neq f(x))$

$$D(A) = D\Big(\{ x \in \mathcal{X} : h(x) \neq f(x) \} \Big) = \mathbb{P}_{x \sim D}\big[h(x) \neq f(x) \big]$$

$$(*) \quad \boxed{=: L_{D,f}(h)} \quad \text{generalization error}$$

The <u>empirical</u> version of (*) is

$$\frac{1}{N} \cdot \left| \left\{ i \in \{1, \dots, N\} : h(x_i) \neq f(x_i) \right\} \right| =: \boxed{L_S(h)}$$

We call this the empirical error/risk.

(<u>Notation</u>: $[N] = \{1, \dots, N\}$
$S|_X = (x_1, \dots, x_N)$)

We can also write the empirical error as

$$L_S(h) = \frac{1}{N} \cdot \sum_{i=1}^{N} \mathbb{1}_{h(x_i) \neq f(x_i)} \quad \text{with}$$

$$\mathbb{1}_{h(x) \neq f(x)} = \begin{cases} 1, & \text{if } h(x) \neq f(x) \\ 0, & \text{else} \end{cases}$$

<u>Claim</u> (relating L_S to $L_{\mathcal{D},f}$):

$$\mathop{\mathbb{E}}_{S|_X \sim \mathcal{D}^N} \left[L_S(h) \right] = L_{\mathcal{D},f}(h)$$

<u>Proof</u>: next week!

03/10/26

$$\mathbb{E}_{S|_x \sim D^m}[L_S(h)] = \mathbb{E}_{S|_x \sim D^m}\left[\frac{1}{N}\cdot\sum_{i=1}^{N} \mathbb{1}_{h(x_i)\neq f(x_i)}\right] \quad // \text{by def.}$$

$$= \frac{1}{N}\cdot\sum_{i=1}^{N} \mathbb{E}_{x_i\sim D}\left[\mathbb{1}_{h(x_i)\neq f(x_i)}\right] \quad // \text{by linearity of } \mathbb{E}[\cdot]$$

$$= \frac{1}{N}\cdot\sum_{i=1}^{N} \mathbb{E}_{x\sim D}\left[\underbrace{\mathbb{1}_{h(x)\neq f(x)}}_{\text{only takes values } 0 \text{ or } 1}\right] \quad // \text{as all } x_i\text{'s are iid}$$

$$= \frac{1}{N}\cdot\sum_{i=1}^{N} \mathbb{P}_{x\sim D}\left[h(x)\neq f(x)\right]$$

$$= \frac{1}{\cancel{N}}\cdot\cancel{N}\cdot\mathbb{P}\left[h(x)\neq f(x)\right]$$

$$= \mathbb{P}_{x\sim D}\left[h(x)\neq f(x)\right]$$

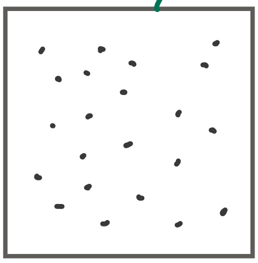
$$= L_{D,f}(h) \quad // \text{by def.}$$

Our first learning paradigm

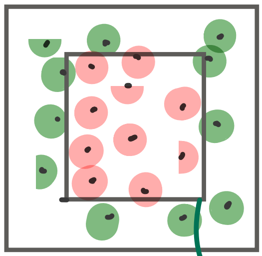
Empirical risk minimization (ERM): As a learner only has access to S (our training data), it is "natural" to try to select h (our hypothesis) such that the empirical risk/error $L_S(h)$ is minimized. We call such a hypothesis an empirical risk minimizer (h_S)!

Example:

we assume the area is 2



Domain $\mathcal{X}$



label 0, label 1 } given by f

we assume the area is 1 (our true labeling function f)

The distribution on $\mathcal{X}$ is a uniform distribution.

Lets say we have an algorithm that returns h_S

$$h_S(x) = \begin{cases} y_i, & \text{if } \exists i \in \{1, \dots, N\} : x_i = x \\ 0, & \text{else} \end{cases}$$

Clearly, h_S is correct on all training data points in $S \Rightarrow h_S$ is an empirical risk minimizer.

$$L_S(h_S) = 0 \,!$$

But, on unseen data points drawn from $\mathcal{D}$ (which is a uniform distribution on $\mathcal{X}$), h_S is only correct 50% of the time $\Rightarrow$

$$L_{\mathcal{D},f}(h_S) = \frac{1}{2} \,!$$

That's what we call over fitting!!!

Hypothesis class (H): We restrict searching for h to H, i.e., a class of functions from $\mathcal{X} \to \mathcal{Y}$ and we write

$$\boxed{ERM_H(S) \in \underset{h \in H}{\operatorname{argmin}}\, L_S(h)}$$

ERM over finite hypothesis classes ($|H| < \infty$)

Assumption A.3 : $\exists h^* \in H$ with $L_{D,f}(h^*) = 0$!
(realizability)

Now, any ERM hypothesis h_S will attain 0 empirical error (meaning $L_S(h_S) = 0$!), as h_S competes against h^* and h^* has $L_S(h^*) = 0$ (and also $L_{D,f}(h^*) = 0$ by A.3).

We know that $L_{D,f}(h_S) > \varepsilon$, $\varepsilon \in (0,1)$, can only happen if our learner selects a hypothesis h_S with $L_S(h_S) = 0$, but $L_{D,f}(h_S) > \varepsilon$.

We define

$$H_{BAD} = \{ h \in H : L_{D,f}(h) > \varepsilon \}$$

(set of BAD hypotheses)

We also define

$$M = \{ S|_x : \exists h \in H_{BAD} ,\ L_S(h) = 0 \}$$

("misleading" training data sets)

<u>Observation:</u>

ERM hypothesis

$$\{ S|_x : L_{D,f}(h_S) \geq \varepsilon \} \subseteq \{ S|_x : \exists h \in H_{BAD} ,\ L_S(h) = 0 \} = M$$

Since

$$\{ S|_x : \exists h \in H_{BAD} ,\ L_S(h) = 0 \} = \bigcup_{h \in H_{BAD}} \{ S|_x : L_S(h) = 0 \}$$

we have

$$D^N \left(\{ S|_x : \exists h \in H_{BAD} ,\ L_S(h) = 0 \} \right) = D^N \left(\bigcup_{h \in H_{BAD}} \{ S|_x : L_S(h) = 0 \} \right)$$

$$\leq \sum_{h \in H_{BAD}} D^N \left(\{ S|_x : L_S(h) = 0 \} \right)$$

(by the "union bound")

Let us fix one $h \in H_{BAD}$ for a moment!

$$\underbrace{\mathcal{D}^N\left(\{S|_x : L_S(h)=0\}\right)}_{\text{one of the summation terms}} = \mathcal{D}^N\left(\{S|_x : \forall i \in \{1,\dots,N\} : h(x_i)=f(x_i)\}\right)$$

$$= \prod_{i=1}^{N} \mathcal{D}\left(\{x \in \mathcal{X} : h(x) = f(x)\}\right)$$

(by our iid assumption A.1)

Remember that $h \in H_{BAD}$, so $L_{\mathcal{D},f}(h) > \varepsilon$.

$$= \prod_{i=1}^{N} \left(1 - \underbrace{L_{\mathcal{D},f}(h)}_{>\varepsilon}\right)$$

$$\leq \prod_{i=1}^{N} (1-\varepsilon)$$

$$= (1-\varepsilon)^N$$

$$\leq e^{-\varepsilon N} \quad \text{(next week)}$$

Overall, we have

$$\mathcal{D}^N\left(\left\{S|_x : L_{\mathcal{D},f}(h_S) > \varepsilon\right\}\right) \leq \sum_{h \in H_{BAD}} \mathcal{D}^N\left(\left\{S|_x : L_S(h) = 0\right\}\right)$$

ERM hyp.

$$\leq \sum_{h \in H_{BAD}} e^{-\varepsilon N} \quad \text{(from before)}$$

$$\leq |H_{BAD}| \cdot e^{-\varepsilon N}$$

$$\leq |H| \cdot e^{-\varepsilon N} \quad \text{since } H_{BAD} \subseteq H$$

If we want $|H| \cdot e^{-\varepsilon N}$ to be less than $\delta \in (0,1)$, we can solve for N, i.e.,

$$|H| \cdot e^{-\varepsilon N} < \delta$$

$$\Rightarrow \boxed{N > \frac{1}{\varepsilon} \cdot \log\left(\frac{|H|}{\delta}\right)}$$

error $\qquad$ confidence $(1-\delta)$

This means, the probability of the generalization error of an ERM hypothesis h_S being larger or equal to $\varepsilon \in (0,1)$ is upper-bounded by $|H| \cdot e^{-\varepsilon N}$ (to be understood as the probability of a random draw of a dataset of size N, iid from the distribution $\mathcal{D}$).

Remark: MARKOV inequality

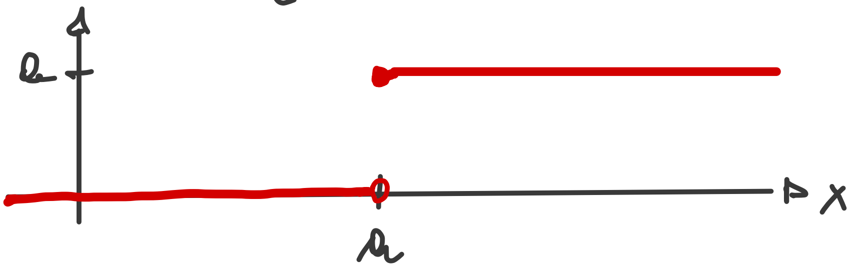
We have a probability space $(S, \mathcal{F}, \mathbb{D})$ and a non-negative random variable X $(X \geq 0)$.

Claim: for all $a > 0$, it holds that

$$\boxed{\mathbb{P}[X \geq a] \leq \frac{\mathbb{E}[X]}{a}}$$

Proof: define $\varphi : S \to \mathbb{R}$

$$\varphi(x) = \begin{cases} a, & \text{if } x \geq a \\ 0, & \text{else} \end{cases}$$



We have that

$$0 \leq \varphi(X(x)) \leq X(x) \quad \text{for all } x \geq 0$$

$$\Rightarrow \int X \, d\mathbb{D} \geq \underbrace{\int \varphi(X(x)) \, d\mathbb{D}}_{a \cdot \underbrace{\mathbb{D}(\{\omega \in S : X(\omega) \geq a\})}_{\mathbb{P}[X \geq a]}} \quad \text{(by monotonicity)}$$

Since $a > 0$: $\frac{1}{a} \cdot \underbrace{\int X \, d\mathbb{D}}_{\mathbb{E}[X]} \geq \mathbb{P}[X \geq a] \Rightarrow \mathbb{P}[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$ $\blacksquare$

Corollary: Let $|H| < \infty$ (finite) and $\varepsilon, \delta \in (0,1)$. Further, let N be an integer such that $N \geq \frac{1}{\varepsilon} \cdot \log\left(\frac{|H|}{\delta}\right)$. Then, for any labeling function $f : \mathcal{X} \to \{0,1\}$ and any distribution $\mathcal{D}$ (for which realizability holds), we have that with probability of at least $1-\delta$ over the choice of $S|_x$ of size N, every ERM hypothesis h_S satisfies

$$L_{\mathcal{D},f}(h_S) \leq \varepsilon.$$

Interpretation: For sufficiently large N, ERM_H returns a hypothesis h_S that is

PROBABLY APPROXIMATELY CORRECT (PAC).

Def. (PAC learnability): A hypothesis class H is PAC learnable if there exists a function $m_H : (0,1)^2 \to \mathbb{N}$ and a learning algorithm A with the following properties: (I) for every $\varepsilon, \delta \in (0,1)$ and (II) every distribution $\mathcal{D}$ over domain $\mathcal{X}$, and (III) every (true) labeling function $f : \mathcal{X} \to \{0,1\}$, if (IV) realizability holds, then running A on $N \geq m_H(\varepsilon, \delta)$ iid instances from $\mathcal{D}$ (labeled by f) returns h_S such that with probability of at least $1-\delta$ (over $S|_x$), it holds that $L_{\mathcal{D},f}(h_S) \leq \varepsilon$.

From now on, we will call our sample size m (not N).

<u>Terminology</u>: $m_H : (0,1)^2 \to \mathbb{N}$ is called the
sample complexity (function).

03/24/2026

In our next step, we are going to remove the realizability assumption. (A.3). This means, there is <u>no longer</u> a $h^* \in H$ with $L_{D,f}(h^*) = 0$. So, the best possible thing we can hope for is a guarantee relative to $\underbrace{\min_{h \in H} L_{D,f}(h)}_{\text{with realizability, this is } 0!!!}$.

<u>Def.</u> (Höffding inequality): Let $X_1, \dots, X_m$ be m iid random variables (RVs) taking values in $[a_i, b_i]$ for $i \in \{1, \dots, m\}$.

Then, given that

$$S_m = \sum_{i=1}^{m} X_i$$

it holds that

$$\text{(a)} \quad \mathbb{P}\left[S_m - \mathbb{E}[S_m] > \varepsilon \right] \leq e^{-\frac{2\varepsilon^2}{\sum_i (b_i - a_i)^2}}$$

$$\text{(b)} \quad \mathbb{P}\left[S_m - \mathbb{E}[S_m] < -\varepsilon \right] \leq \text{— '' —}$$

Upon combination (using the union bound), we get

$$\mathbb{P}\left[\,\left|S_m - \mathbb{E}[S_m]\right| > \varepsilon\right] \leq 2 \cdot e^{-\frac{2\varepsilon^2}{\sum_i (b_i - a_i)^2}}$$

(without proof!)

For us, the following alternative formulation will be helpful:

!

$$\mathbb{P}\left[\,\left|\frac{1}{m}\cdot\sum_{i=1}^{m} X_i - \mu\right| > \varepsilon\right] \leq 2 \cdot e^{-\frac{2\varepsilon^2 m}{(b-a)^2}}$$

with $\mu := \mathbb{E}[X_i]$ and $\mathbb{P}[a \leq X_i \leq b] = 1$ (for $i \in \{1,\dots,m\}$)

By using the Höffding inequality, we can already say the following: fix $\varepsilon > 0$; then, for a <u>single</u> hypothesis $h : X \to Y$, we have

$$\mathbb{P}_{S|x \sim D^m}\left[\,\left|\underbrace{L_S(h)}_{\frac{1}{m}\cdot\sum_{i=1}^{m}\mathbb{1}_{h(x_i)\neq f(x_i)}} - L_{D,f}(h)\right| > \varepsilon\right] \leq 2 \cdot e^{-2\varepsilon^2 m} \quad \left(\text{since } \begin{matrix} b=1 \\ a=0 \end{matrix}\right)$$

Importantly, this only holds for a single h!!!

We actually want a bound that holds uniformly over <u>all</u> hypotheses in the class H. For finite hypothesis classes ($|H| < \infty$), we get such a result by the <u>union bound</u>.

Formally:

for a single h!

$$\Pr_{S|x\sim D^m}\left[\exists h\in H: |L_S(h)-L_{D,f}(h)|>\varepsilon\right] \overset{\text{U.B.}}{\leq} \sum_{h\in H} \overbrace{2\cdot e^{-2\varepsilon^2 m}}$$

$$\leq |H|\cdot 2\cdot e^{-2\varepsilon^2 m}$$

with realizability, we had ε here not ε^2!

We can next also get rid of the true labeling function f very easily by letting D be a distribution over $\mathcal{X}$ and $\mathcal{Y}$, i.e., over the domain $\mathcal{X}$ and label set $\mathcal{Y}$.

Let $Z := X\times Y$. We are going to adapt our definitions for the generalization and empirical error as follows:

$$L_D(h) = \Pr_{(x,y)\sim D}\left[h(x)\neq y\right] = D\left(\left\{(x,y)\in Z : h(x)\neq y\right\}\right)$$

$$L_S(h) = \frac{1}{m}\cdot\left|\left\{i\in\{1,\dots,m\} : h(x_i)\neq y_i\right\}\right|$$

<u>Def.</u> (Agnostic PAC learnability): A hyp. class H of functions $h: X \to Y$ is agnostic PAC learnable if there exists $m_H : (0,1)^2 \to \mathbb{N}$ and an algorithm A with the following properties: (I) for every $\varepsilon, \delta \in (0,1)$ and (II) every distribution D over $Z := X \times Y$, when running A on m iid instances from D with $m \geq m_H(\varepsilon, \delta)$, A returns a hypothesis h such that with probability of at least $1-\delta$ (over the choice of S) it holds that

$$L_D(h) \leq \min_{h' \in H} L_D(h') + \varepsilon$$

<u>Another generalization of setup:</u>

Loss function $\ell : H \times X \times Y \longrightarrow \mathbb{R}_+$

<u>Example:</u> "0-1 loss"

$$\ell^{0-1}(h,(x,y)) = \begin{cases} 1, \text{ if } h(x) \neq y \\ 0, \text{ if } h(x) = y \end{cases}$$

With this definition, we can write

$$L_D(h) = \mathop{\mathbb{E}}_{\substack{z \sim D \\ (x,y)}} \left[\ell^{0-1}(h,z)\right] = 1 \cdot \mathbb{P}\left[h(x) \neq y\right] + \cancel{0 \cdot \mathbb{P}\left[h(x) = y\right]}$$

$$= \mathop{\mathbb{P}}_{(x,y) \sim D}\left[h(x) \neq y\right]$$

04/14/26

Uniform Convergence

Def. (ε-representative sample): A sample S is called ε-representative with respect to $Z = X \times Y$, hypothesis class H, loss function ℓ, and distribution D, if

$$\boxed{\forall h \in H: \quad |L_S(h) - L_D(h)| \leq \varepsilon}$$

Lemma: Assume that S is $\frac{\varepsilon}{2}$-representative wrt. Z, H, ℓ and D.
Then, any hypothesis h_S returned by

$$ERM_H(S) \in \operatorname*{argmin}_{h' \in H} L_S(h')$$

satisfies

$$L_D(h_S) \leq \min_{h \in H} L_D(h) + \varepsilon$$

Proof: for any $h \in H$:

$$\begin{aligned}
L_D(h_S) &\leq L_S(h_S) + \varepsilon/2 && \text{// by } \tfrac{\varepsilon}{2}\text{-representativeness} \\
&\leq L_S(h) + \tfrac{\varepsilon}{2} && \text{// since } h_S \text{ is an ERM hyp.} \\
&\leq L_D(h) + \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} && \text{// by } \tfrac{\varepsilon}{2}\text{-representativeness} \\
&= L_D(h) + \varepsilon && (*)
\end{aligned}$$

We conclude: since (*) holds for any $h \in H$, we get

$$L_D(h_S) \leq \min_{h\in H} L_D(h) + \varepsilon$$

__Def.:__ (uniform convergence (UC)): A hypothesis class H has the UC property with respect to $Z = X \times Y$ and loss function ℓ, if there exists $m_H^{uc} : (0,1)^2 \to \mathbb{N}$ such that for any $\varepsilon, \delta \in (0,1)$ and every distribution D over $Z = X \times Y$, if S is an iid sample of size $m \geq m_H^{uc}(\varepsilon,\delta)$ from D, then with probability of at least $1-\delta$ (over the choice of S), S is __ε-representative.__

__Corollary:__ If H satisfies the UC property with sample complexity function m_H^{uc}, then H is __agnostic PAC learnable__ with

$$\underbrace{m_H(\varepsilon,\delta)}_{\text{sample complexity for agnostic PAC}} \leq m_H^{uc}\left(\tfrac{\varepsilon}{2},\delta\right)$$

by ERM (i.e., our learning paradigm).

Question: Is there a universal learner? By universal we mean a learner without any prior knowledge of the learning task/problem (given by $\mathcal{D}$), but can be challenged by any task and still returns $A(S)$ with low $L_{\mathcal{D}}(A(S))$.

Theorem (No-Free-Lunch): Let A be a learning algorithm for the task of binary classification with respect to the 0-1 loss, over domain $\mathcal{X}$. Also let m be any number ($\in \mathbb{N}$) smaller than $|\mathcal{X}|/2$ (representing the size of S). Then, there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \underbrace{\{0,1\}}_{\mathcal{Y}}$ such that

1. $\exists f: \mathcal{X} \to \{0,1\}$ with $L_{\mathcal{D}}(f) = 0$!

and

2. with probability of at least $1/7$ over the choice of $S \sim \mathcal{D}^m$, we have

$$L_{\mathcal{D}}(A(S)) \geq 1/8.$$

Interpretation: (1) means that the task can be learned by another learner (with, e.g., $H = \{f\}$), and (2) means that A fails on that task!

Corollary: Let $\mathcal{X}$ be infinitely large and H be the class of **all** functions from $X \to \{0,1\}$, then H is **not** PAC learnable.

VAPNIK - CHERVONENKIS (VC) - Dimension

Def: Let H be a class of functions from $X \to \{0,1\}$ and let $C \subset \mathcal{X}$, $C = \{c_1, \dots, c_m\}$. We define

$$H_C = \{(h(c_1), h(c_2), \dots, h(c_m)) : h \in H\}$$

as the restriction of H to the subset C.

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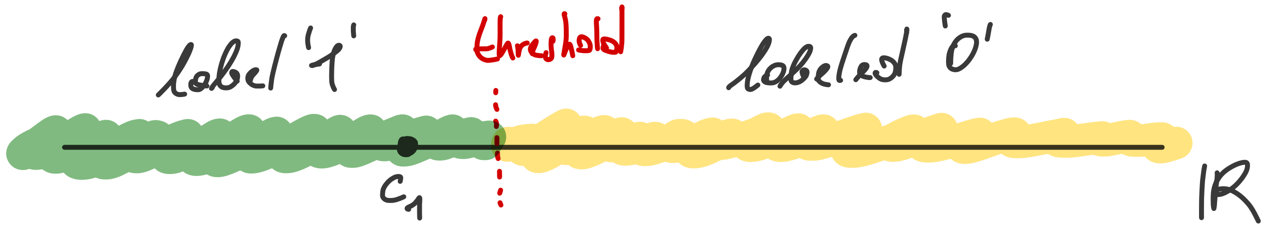
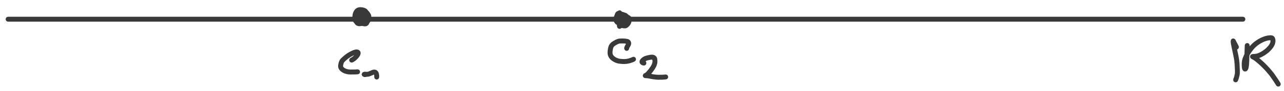
Example: $C = \{c_1\}$, $c_1 \in \mathbb{R}$ and let H^{thr} be the class of thresholds on the real line $\mathbb{R}$.



$$H_C^{thr} = \{(1), (0)\} \Rightarrow |H_C^{thr}| = 2 = 2^1$$

Lets take $C = \{c_1, c_2\}$; without loss of generality we say $c_1 \leq c_2$



We get

$$H_C^{thr} = \{(1,1), (1,0), (0,0)\}$$

$$\Rightarrow |H_C^{thr}| = 3$$

But, we would have $2^2 = 4$ <u>possible</u> labelings of $C = \{c_1, c_2\}$.

<u>Def.</u> (Shattering): A hypothesis class H shatters a finite set C of size m if $|H_C| = 2^m$.

<u>Claim:</u> The class H^{thr} on $\mathcal{X} = \mathbb{R}$ is PAC-learnable with

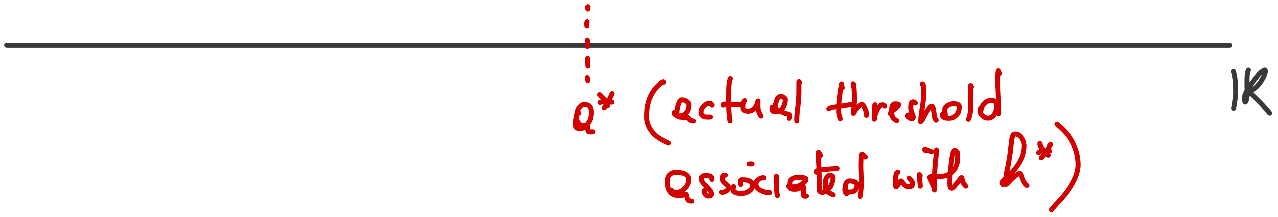
$$\boxed{m_{H^{thr}}(\varepsilon, \delta) \leq \left\lceil \frac{1}{\varepsilon} \cdot \log\left(\frac{2}{\delta}\right) \right\rceil}$$

$$H^{thr} = \{h_a : a \in \mathbb{R}\}, \quad h_a(x) = \begin{cases} 1, & \text{if } x < a \\ 0, & \text{else} \end{cases}$$

threshold

(or $h_a(x) = \mathbb{1}_{x < a}$)

<u>Proof:</u> We assume <u>realizability</u>, i.e., $\exists h^* \in H^{thr}$ such that $L_{D,f}(h^*) = 0$.



We let $a_0 \in \mathbb{R}$ be such that

$$D(\{x \in \mathbb{R} : x \in (a_0, a^*)\}) = \varepsilon$$

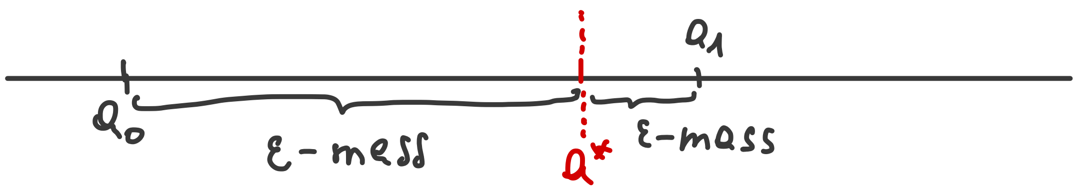
and let $a_1 \in \mathbb{R}$ be such that

$$D(\{x \in \mathbb{R} : x \in (a^*, a_1)\}) = \varepsilon$$

<u>Special cases:</u>

In case $D(\{x \in \mathbb{R} : x \in (a_0, a^*)\}) < \varepsilon$, then set $a_0 = -\infty$

In case $D(\{x \in \mathbb{R} : x \in (a^*, a_1)\}) < \varepsilon$, then set $a_1 = +\infty$



Next, we are going to define an <u>ERM algorithm</u> (needed for PAC-learnability).

(1) Pick $b_0 \in \mathbb{R}$ and $b_1 \in \mathbb{R}$ from $S = \big((x_1, y_1), \ldots, (x_m, y_m)\big)$

as

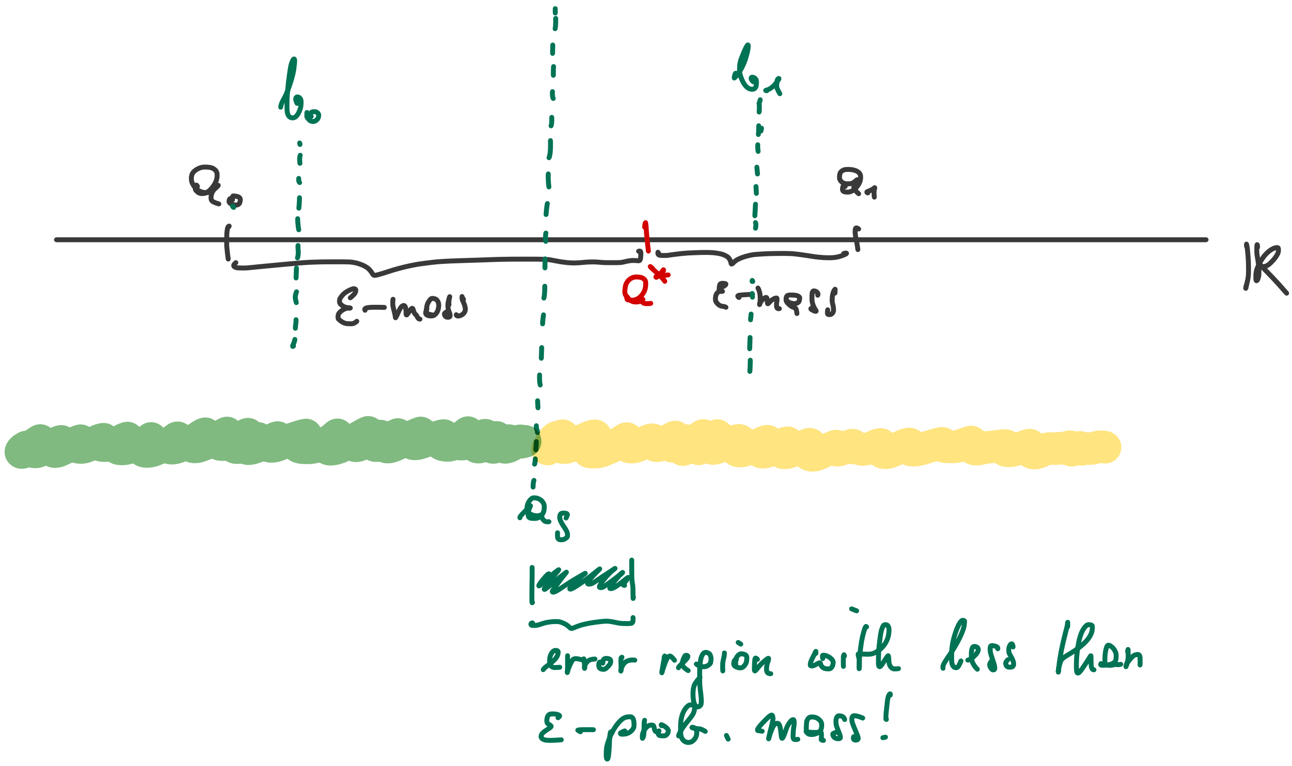
$$b_0 = \max \{x : (x,1) \in S\}$$
$$b_1 = \min \{x : (x,0) \in S\}$$

(2) Pick <u>any</u> threshold within (b_0, b_1) and call the corresponding hypothesis h_S.

We see that h_S always has 0 empirical error !!!

By our construction of a_0 and a_1, the claim is that for h_S (our ERM hypothesis) to have $L_{D,f}(h_S) \leq \varepsilon$, it suffices that

$$\boxed{\begin{array}{ll} (a) & b_1 \leq a_1 \quad \underline{\text{and}} \\ (b) & b_0 \geq a_0 \end{array}} \qquad (*)$$



Formally, we have

$$\mathbb{P}\big[L_{D,f}(\underset{\substack{\uparrow \\ \text{ERM hyp.}}}{h_S}) > \varepsilon\big] \leq \mathbb{P}\big[(b_0 < a_0) \ \underline{\text{or}}\ (b_1 > a_1)\big]$$

$$\overset{\text{U.B.}}{\leq} \mathbb{P}\big[b_0 < a_0\big] + \mathbb{P}\big[b_1 > a_1\big]$$

To upper-bound $\mathbb{P}[L_{D,f}(h_S) > \varepsilon]$, we need to check when $\underline{b_0 < a_0}$ (or $b_1 > a_1$). This only happens when there is $\underline{\text{no}}$ instance x in S that is labeled 1 such that $x \in (a_0, a^*)$.

We know that $D(\{x \in \mathbb{R} : x \in (a_0, a^*)\}) = \varepsilon$ (by construction). Hence, <u>not seeing</u> an instance in (a_0, a^*) has probability of $1-\varepsilon$, and not seeing an instance in m iid samples has probability of $(1-\varepsilon)^m$.

Overall: $\mathbb{P}\left[L_{D,f}(h_S) > \varepsilon\right] \leq 2\cdot(1-\varepsilon)^m$

$$\leq 2e^{-\varepsilon m}$$

If we want $2e^{-\varepsilon m} < \delta$ for $\delta \in (0,1)$, we get

$$\boxed{m > \frac{1}{\varepsilon}\cdot \log\left(\frac{2}{\delta}\right)}$$

We have seen that finiteness of H is sufficient, but not necessary for PAC learnability.

05/05/26

Def. (VC-Dimension): The VC-Dimension of H (i.e., a class of functions from $X \to \{0,1\}$), written as

$$VC(H)$$

is the maximal size of a set $C \subset X$ that is shattered by H.

Theorem: Let H be a class of functions from $X \to \{0,1\}$. if H has infinite VC-Dimension, then H is not PAC learnable.

Def. (growth function): Let H be a class of functions from $X \to \{0,1\}$. The growth function of H, $\tau_H : \mathbb{N}_0 \to \mathbb{N}_0$, is defined as

$$\tau_H(m) = \max_{\substack{C \subset X \\ |C| = m}} |H_C|$$

convention is $\tau_H(0) = 1$

(for H_{thr} on the real line
$\tau_{H_{thr}}(0) = 1$, $\tau_{H_{thr}}(1) = 2$, $\tau_{H_{thr}}(2) = 3$, ...)
$(2^m = 2^1 = 2)$

Lemma (Sauer, Shelah, Perez, "Sauer's Lemma"): Let H be a class of functions from $X \to \{0,1\}$ with $VC(H) = d < \infty$.

Then

$$\tau_H(m) = 2^m \text{ if } m \leq d, \text{ BUT}$$
$$\tau_H(m) \leq \left(\frac{e \cdot m}{d}\right)^d \text{ if } m > d$$

(If we take any set C (of size m) and $\underbrace{|H| < \infty}_{\text{finite class}}$, we know

$$|H_C| \leq |H|$$

So, if $|H| < 2^m$, then obviously H <u>can not</u> shatter C of size m. This implies

$$VC(H) \leq \log_2 (|H|)\text{)}$$

<u>Theorem</u>: Let H be a class of functions from $X \to \{0,1\}$ and $\ell : H \times X \times Y \to [0, c]$, $c > 0$, a loss function. For any distribution D over $X \times Y$ and $\delta \in (0,1)$, we have with probability of at least $1-\delta$ (over the choice of $S \sim D^m$)

$$\forall h \in H : \left| L_D(h) - L_S(h) \right| \leq c \cdot \sqrt{\frac{8 \cdot \log\left(\tau_H(2m) \cdot \frac{4}{\delta}\right)}{m}}$$

Proof: Take $S \sim D^m$ and $S' \sim D^m$.

Lets look at $|L_{S'}(h) - L_S(h)|$. First, we expand the terms as follows:

$$\left|\left(L_D(h) - L_S(h)\right) - \left(L_D(h) - L_{S'}(h)\right)\right| \geq \left|L_D(h) - L_S(h)\right| - \left|L_D(h) - L_{S'}(h)\right|$$

(reverse triangle ineq.)

if (1) $|L_D(h) - L_S(h)| > \varepsilon$ and

(2) $|L_D(h) - L_{S'}(h)| < \frac{\varepsilon}{2}$, then $|L_{S'}(h) - L_S(h)| > \frac{\varepsilon}{2}$

Rewritten in terms of indicator functions:

$$\mathbb{1}_{|L_D(h) - L_S(h)| > \varepsilon} \cdot \mathbb{1}_{|L_D(h) - L_{S'}(h)| < \frac{\varepsilon}{2}} \leq \mathbb{1}_{|L_{S'}(h) - L_S(h)| > \frac{\varepsilon}{2}}$$

Lets take the expectation over S':

$$\boxed{1} \quad \mathbb{E}_{S'}\left[\mathbb{1}_{|L_D(h) - L_{S'}(h)| < \frac{\varepsilon}{2}}\right] = \mathbb{P}_{S'}\left[|L_D(h) - L_{S'}(h)| < \frac{\varepsilon}{2}\right]$$

by Höffding ineq. $$\geq 1 - 2e^{-2\cdot\left(\frac{\varepsilon}{2}\right)^2 m^2 / \sum_i (c-0)^2}$$

$$= 1 - 2e^{-\frac{\varepsilon^2 m}{2c^2}} \quad (\times)$$

$\boxed{2}$ $\mathbb{E}_{S'}\left[\mathbb{1}_{|L_{S'}(h)-L_S(h)|>\frac{\varepsilon}{2}}\right] = \mathbb{P}_{S'}\left[|L_{S'}(h)-L_S(h)|>\frac{\varepsilon}{2}\right]$

<u>Question:</u> Can we further bound (x) $1-2e^{-\varepsilon^2 m/2c^2}$

05/19/2026

1st lecture exam date = last lecture.

<u>Proof continued:</u>

if $m \geq 4c^2\varepsilon^{-2}\cdot\log(2)$ (we have to check this later)

we would get

$$1-2e^{-\varepsilon^2 m/2c^2} \geq 1-2e^{-\cancel{\varepsilon^2} 4\cancel{c^2}\cancel{\varepsilon^{-2}}\log(2)/2\cancel{c^2}}$$

$$= 1-2e^{-2\log(2)}$$

$$= 1-2\cdot\frac{1}{4} = \frac{1}{2}$$

At the moment, we have

$$\mathbb{1}_{|L_D(h)-L_S(h)|>\varepsilon}\cdot\frac{1}{2} \leq \mathbb{E}_{S'}\left[\mathbb{1}_{|L_{S'}(h)-L_S(h)|>\frac{\varepsilon}{2}}\right]$$

$$\Leftrightarrow \quad \mathbb{1}_{|L_D(h)-L_S(h)|>\varepsilon} \leq 2\cdot\mathbb{P}_{S'}\left[|L_{S'}(h)-L_S(h)|>\frac{\varepsilon}{2}\right]$$

$$\leq 2\cdot\mathbb{P}_{S'}\left[\exists h\in H: |L_{S'}(h)-L_S(h)|>\frac{\varepsilon}{2}\right]$$

We can re-write this as

$$\mathbb{1}_{\exists h \in H : |L_D(h) - L_S(h)| > \varepsilon} \leq 2 \cdot \mathbb{P}_{S'}\left[|L_{S'}(h) - L_S(h)| > \frac{\varepsilon}{2} \right]$$

(as $\mathbb{1}_{|L_D(h) - L_S(h)| > \varepsilon}$ is either 0 or 1). So, if we now take the expectation with respect to S, we get

$$\mathbb{E}_S\left[\mathbb{1}_{\exists h \in H : |L_D(h) - L_S(h)| > \varepsilon} \right] \leq 2 \cdot \mathbb{E}_{S, S'}\left[|L_{S'}(h) - L_S(h)| > \frac{\varepsilon}{2} \right]$$

Remember that $L_S(h) = \frac{1}{m} \cdot \sum_{i=1}^{m} \ell(h, z_i)$ and

$$L_{S'}(h) = \frac{1}{m} \cdot \sum_{i=1}^{m} \ell(h, z_i')$$

Hence,

$$|L_{S'}(h) - L_S(h)| = \frac{1}{m} \cdot \left| \sum_{i=1}^{m} \left(\ell(h, z_i') - \ell(h, z_i) \right) \right|$$

As all z_i and z_i' are drawn iid from the distribution D, we can switch them. What happens to

$$\ell(h, z_i) - \ell(h, z_i') \; ?$$

We get a sign flip in the sum!

We can let $\sigma = (+1, -1, +1, \ldots)$ of length m and write

$$\frac{1}{m}\left|\sum_{i=1}^{m} \sigma_i \cdot \left(\ell(h,z_i') - \ell(h,z_i)\right)\right|$$

(the σ_i's are called RADEMACHER variables). We can thus draw σ_i from a uniform distribution on $\{+1,-1\}$, $U(\{\pm 1\})$, take the expectation w.r.t. σ and still <u>not</u> change the distribution of $\ell(h,z_i') - \ell(h,z_i)$.

Formally,

$$\mathbb{E}_S\left[\mathbb{1}_{\exists h \in H : |L_D(h) - L_S(h)| > \varepsilon}\right]$$

$$\leq 2 \cdot \mathbb{E}_{S,S'}\left[\mathbb{P}_{\sigma \sim U(\{\pm 1\})^m}\left[\exists h \in H : \frac{1}{m} \cdot \left|\sum_{i=1}^{m} \sigma_i \overbrace{\left(\ell(h,z_i') - \ell(h,z_i)\right)}^{\text{range is } [0,c]}\right| > \frac{\varepsilon}{2}\right]\right]$$

<u>Lets fix S, S' and $h \in H$:</u> let $\theta_h = \frac{1}{m} \cdot \sum_{i=1}^{m} \sigma_i \left(\ell(h,z_i') - \ell(h,z_i)\right)$

We know $\mathbb{E}_\sigma[\theta_h] = 0$, since $\left(\ell(h,z_i') - \ell(h,z_i)\right)$ is constant (as we fixed h).

This means, we can write

$$\mathbb{P}_{\sigma}\left[\left|\Theta_h - \underbrace{\mathbb{E}[\Theta_h]}_{0}\right| > \frac{\varepsilon}{2}\right] \underset{\text{Höffding}}{\leq} 2 \cdot e^{-2m^2\left(\frac{\varepsilon}{2}\right)^2 / \sum_i \underbrace{(c-(-c))^2}}$$

because the range of Θ_h is $[-c,c]$

$$= 2 \cdot e^{-m\varepsilon^2/8c^2}$$

We can now account for "$\exists h \in H$" by the union bound. This works as $C = S \cup S'$ contains at most $2m$ instances. So, effectively we have H_C and not H anymore.

06/02/26

So, we now have

$$\mathbb{P}_{\sigma}\left[\exists h \in H_C : \left|\frac{1}{m} \cdot \sum_{i=1}^{m} \sigma_i \left(\ell(h, z_i') - \ell(h, z_i)\right)\right| > \frac{\varepsilon}{2}\right] \underset{\text{union bound}}{\leq}$$

$$\underbrace{\tau_H(2m)} \cdot 2 \cdot e^{-m\varepsilon^2/8c^2}$$

from the growth function, we know how many labelings are possible, namely $\tau_H(2m)$

Overall, we get

$$\mathbb{P}_S\left[\exists h \in H : |L_D(h) - L_S(h)| > \varepsilon\right] \leq 4 \cdot \tau_H(2m) \cdot e^{-m\varepsilon^2/8c^2}$$

To get the result from the theorem, we set

$$4\,\tau_H(2m) \cdot e^{-m\varepsilon^2/8c^2} = \delta$$

$$e^{-m\varepsilon^2/8c^2} = \frac{\delta}{4 \cdot \tau_H(2m)}$$

$$\varepsilon^2 = \frac{\log\left(\frac{4\,\tau_H(2m)}{\delta}\right) \cdot 8c^2}{m}$$

$$\longrightarrow \quad \varepsilon = c \cdot \sqrt{\frac{8 \cdot \log\left(\frac{4\,\tau_H(2m)}{\delta}\right)}{m}}$$

The assumption that we made, i.e.,

$$m \geq 4c^2 \varepsilon^{-2} \log(2)$$

is equivalent to $\boxed{\delta \leq 2 \cdot \sqrt{2} \cdot \tau_H(2m)}$

Always fulfilled for $\delta \in (0,1)$

We can now formulate the **fundamental theorem of statistical learning**:

Theorem: Let H be a hypothesis class of functions from X to $\{0,1\}$ and let ℓ be the 0-1 loss. Then, the following statements are equivalent:

$\boxed{1}$ H has the uniform convergence property (UC)

$\boxed{2}$ Any ERM algorithm is a successful agnostic PAC learner for H

$\boxed{3}$ H is agnostic PAC learnable

$\boxed{4}$ H is PAC learnable

$\boxed{5}$ Any ERM algorithm is a successful PAC learner for H

$\boxed{6}$ H has finite VC dimension.

Let's look at some hypothesis classes.

Linear predictors

Define

$$L_d = \{ h_{w,b} : w \in \mathbb{R}^d, b \in \mathbb{R} \}$$

$$h_{w,b}(x) = \langle w, x \rangle + b$$

Class of affine functions

From L_d, we can get different predictors via composition with some $\phi : \mathbb{R} \to Y$. Further, we can use

$w' = (b, w_1, \dots, w_d)$ and

$x' = (1, x_1, \dots, x_d)$ to write

$$h_{w,b}(x) = \langle w', x' \rangle$$

$\Rightarrow$ Affine functions in $\mathbb{R}^d$ can be written as homogeneous linear functions in $\mathbb{R}^{d+1}$.

> if $\phi : \mathbb{R} \to Y$ is $\phi = \text{sign}$
> we get so called HALFSPACE classifiers HS_d.

$$HS_d = \phi \circ L_d = \{ x \mapsto \mathrm{sign}(h_{w,b}(x)),\ h_{w,b} \in L_d \}$$

<u>Question:</u> What is $VC(HS_d)$?

<u>Theorem:</u> The VC dimension of

$$\mathcal{F} = \{ x \mapsto \mathrm{sign}(\langle w, x\rangle) : w \in \mathbb{R}^d \}$$

is d.

(Remark: $\mathcal{Y} = \{\pm 1\}$)

<u>Proof:</u> We first show $VC(\mathcal{F}) \geq d$ (lower bound)

Let $C = \{c_1, \dots, c_d\}$ where $c_i \in \mathbb{R}^d$

$$c_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},\ c_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \dots\dots\ c_d = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Let $w = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{pmatrix}$ for any given labeling (in $\{\pm 1\}$) of $\{c_1, \dots, c_d\}$

j-th coordinate of c_i

$$\Rightarrow \langle w, c_i \rangle = \sum_{j=1}^{d} w_j \cdot \underbrace{c_{ij}}_{1 \text{ at } i=j} = w_i = y_i$$

$$\Rightarrow \boxed{VC(\mathcal{F}) \geq d}$$

06/08/2026

To establish $VC(\mathcal{F}) = d$, we need to show
$$VC(\mathcal{F}) < d+1.$$

We will show this by <u>contradiction</u>.

! Assume $d+1$ points $C = \{c_1, \dots, c_{d+1}\}$ are shattered by $\mathcal{F}$.

This means that $\exists\, w_1, \dots, w_{2^{d+1}}$ weight vectors that will yield <u>all</u> 2^{d+1} possible labelings of C.

Lets write down <u>all</u> possible inner products:

$$M = \begin{array}{c} \xrightarrow{\quad 2^{d+1} \text{ columns} \quad} \\ \begin{pmatrix} w_1^T c_1 & w_2^T c_1 & \cdots & w_{2^{d+1}}^T c_1 \\ w_1^T c_2 & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ w_1^T c_{d+1} & \cdots & \cdots & w_{2^{d+1}}^T c_{d+1} \end{pmatrix} \end{array} \Big\downarrow d+1 \text{ rows}$$

Lets take

$$X = \begin{array}{c} \xrightarrow{\quad d \text{ cols} \quad} \\ \begin{pmatrix} - & c_1 & - \\ - & c_2 & - \\ & \vdots & \\ - & c_{d+1} & - \end{pmatrix} \end{array} \Big\downarrow d+1 \text{ rows}$$

and

$$W = \begin{array}{c} \begin{pmatrix} | & | & & | \\ w_1 & w_2 & \cdots\cdots & w_{2^{d+1}} \\ | & | & & | \end{pmatrix} \Big\downarrow \, d \text{ rows} \\ \xrightarrow{\hspace{6em}} \\ 2^{d+1} \text{ columns} \end{array}$$

We can now write

$$\boxed{M = XW}$$

<u>Remark</u>: if we would take $\text{sign}(XW)$, we would get <u>all</u> possible labelings.

We know that

$$\boxed{\text{rank}(M) \leq \min\{\text{rank}(X), \text{rank}(W)\} = d}$$

$$M = \begin{pmatrix} w_1^T c_1 & \cdots & w_{2d+1}^T c_1 \\ w_1^T c_2 & \ddots & \vdots \\ \vdots & & \vdots \\ w_1^T c_{d+1} & \cdots & w_{2d+1}^T c_{d+1} \end{pmatrix}$$

$X w_1 \in \mathbb{R}^{d+1}$

Any $v_1, \ldots, v_k$ (set of vectors) are <u>linearly independent</u> if and only if $(*)$

$$a_1 \cdot v_1 + a_2 \cdot v_2 + \cdots + a_k v_k = 0$$

for $\forall i : a_i = 0$

In our case (for M), if we look at the <u>rows</u> of M

$$a_1 \cdot \begin{bmatrix} w_1^T c_1 & \cdots & w_{2d+1}^T c_1 \end{bmatrix} +$$

$$a_2 \cdot \begin{bmatrix} w_1^T c_2 & \cdots & w_{2d+1}^T c_2 \end{bmatrix} +$$

$$\vdots$$

$$a_{d+1} \begin{bmatrix} w_1^T c_{d+1} & \cdots & w_{2d+1}^T c_{d+1} \end{bmatrix}$$

Letting $a = [a_1, \ldots a_{d+1}]$, we can write this as

$$\left[a^T X w_1 \quad a^T X w_2 \quad \cdots\cdots \quad a^T X w_{2^{d+1}} \right]$$

As we assume that $c_1 \ldots c_{d+1}$ are <u>shattered</u> by $\mathcal{F}$, there (by assumption)

$\exists k \in \{1, \ldots\ldots, 2^{d+1}\}$ such that

$$\text{sign}(a) = \text{sign}(X w_k)$$

This means that

$$a^T X w_k$$

is a sum of <u>positive</u> numbers $\Rightarrow \neq 0$! In turn, by $(\ast)$ this means we have $d+1$ linearly independent rows in M, i.e., $\text{rank}(M) = d+1$.

$\Rightarrow$ contradiction to our earlier statement of $\text{rank}(M) \leq d$!

We conclude that $VC(\mathcal{F}) < d+1$ and thus (with our lower bound)

$$\boxed{VC(\mathcal{F}) = d}$$

$\square$

Boosting

Def. (γ-weak learnability): A learning algorithm A is a γ-weak learner for a hypothesis class H if there exists $m_H : (0,1) \to \mathbb{N}$ such that for every $\delta \in (0,1)$, for every distribution D over X, and for every labeling function $f: X \to \{\pm 1\}$, if realizability holds with respect to H, D, f, then when running A on $m \geq m_H(\delta)$ iid samples from D (labeled by f), the algorithm returns a hypothesis h such that with probability of at least $1-\delta$ (over the choice of m samples from D)

$$L_{D,f}(h) \leq \frac{1}{2} - \gamma$$